

KEERTHANA RAVIKUMAR

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SUMMARY

Full-Stack Software Engineer with 2.5+ years of professional experience building scalable web applications and enterprise-grade backend services. Proficient in developing REST APIs, microservices, and responsive front-end interfaces using Java, Spring Boot, React.js, AWS, and Docker. Experienced in integrating cloud-based services, CI/CD pipelines, and relational/non-relational databases. Currently pursuing a Master's in Computer Software Engineering to deepen software architecture, cloud technologies, and full-stack development expertise. Strong problem-solving skills demonstrated through 250+ coding challenges and collaborative, cross-functional project delivery.

EDUCATION

Arizona State University | *Master's in Computer Software Engineering – 3.8 GPA*

Advanced Algorithms and Data structures, Embedded Systems – Internet of Things, Software Architecture, Software Verification, Validation and Testing.

Tempe, AZ, USA

Jan 2024 - Dec 2025

Visveswaraya Technological University | *Bachelor's in Computer Science and Engineering - 3.4 GPA*

Bengaluru, KA, IN

Aug 2017 – Jul 2021

SKILL SET

TECHNICAL SKILLS

- Languages: Java, Python, JavaScript, C, Kotlin, PHP, SQL
- Frameworks & Libraries: Spring Boot, React.js, React Native, Node.js, Junit
- Databases & Storage: MySQL, PostgreSQL, MongoDB
- Cloud & DevOps: AWS, Docker, Jenkins, CICD, Git, Linux
- Core Concepts: Data Structures & Algorithms, Object-Oriented Programming, System Design, REST APIs, Microservices, Agile/Scrum, Load Testing

LEADERSHIP AND IMPACT

- **Technical Mentorship:** Guided 3 junior developers in Spring Boot development practices and code review processes, contributing to improved team code quality standard
- **Process Improvement:** Introduced automated testing practices and standardized documentation, reducing feature deployment cycle from 2 weeks to 1 week
- **Knowledge Sharing:** Presented technical sessions on microservices best practices and API design patterns to development team of 20+ engineers
- **Stakeholder Communication:** Collaborated with product managers and business stakeholders to translate requirements into technical specifications and project timelines

PROFESSIONAL EXPERIENCE

Mercedes Benz Research and Development India | *Software Engineer*

Aug 2021 – Dec 2023

- **Backend Development:** Developed Spring Boot microservices handling 50,000+ daily API requests with maintained 99%+ uptime through proper error handling and monitoring
- **Feature Implementation:** Contributed to development of 15+ customer-facing features using Java 11, Spring Boot, and MySQL, deployed across multiple European markets
- **Performance Optimization:** Improved API response times by 35% through database query optimization and Redis caching implementation, supporting 2x increase in concurrent users
- **Code Quality Leadership:** Led code review initiatives and established testing standards, resulting in 50% reduction in production defects and improved team development practices
- **Cross-functional Collaboration:** Worked closely with product managers, QA engineers, and DevOps teams in agile environment to deliver features meeting business requirements
- **Internal Tool Development:** Built employee skill tracking application using Spring Boot and Angular, reducing manual HR processes and improving team visibility into technical competencies
- **Documentation & Onboarding:** Created comprehensive technical documentation and onboarding materials, reducing new developer ramp-up time from 3 weeks to 2 weeks

HeadSpin, Inc | *Associate Automation and Test Engineer Intern*

Apr 2021 – July 2021

- **API Testing:** Programmed automated test suites using Java, TestNG, and REST Assured for mobile testing platform, achieving 80%+ test coverage across critical user flows
- **Technical Proof of Concepts:** Contributed to 8 technical demonstrations showcasing mobile performance testing capabilities for prospective enterprise clients
- **Learning & Adaptation:** Gained proficiency in Java Spring framework, REST API testing methodologies, and mobile application testing within 3-month internship period

TECHNICAL PROJECTS

ENCRYPTUS | *HTML, CSS, JavaScript, PHP*

- Coded secure file transfer web application using PHP, MySQL, HTML5, CSS3, and JavaScript with WeTransfer-inspired functionality and end-to-end encryption implementation, achieving 78% increase in secure transaction volume over 6 months
- Designed custom encryption pipeline using PHP OpenSSL library for enhanced data integrity, reducing external security vulnerability reports by 37% through advanced cryptographic protocols
- Implemented token-based validation system with dynamic link expiration and user-authenticated access controls, reducing file misuse and access breaches by 40%
- Built secure file upload and download system with AES-256 encryption frontend implementation and backend secure link generation using PHP OpenSSL library, maintaining zero unauthorized access incidents during pilot phase

FARMATO | *HTML, CSS, JavaScript, PHP, MySQL*

- Engineered bot controlled water level detection and replenishment system using IoT sensors and microcontrollers, eliminating manual monitoring processes and saving 40 labor hours per week through autonomous operation
- Designed user interface application for plant watering schedule optimization using JavaScript, HTML, CSS, and database integration, enhancing operational efficiency by 41% and reducing manual intervention by 25 hours per month
- Deployed intelligent sensor network including temperature sensors, soil moisture sensors, and water level detectors for autonomized plant irrigation system, achieving 32% water usage reduction and 26% improvement in plant health metrics
- Built Internet of Things (IoT) data analytics system for real-time sensor monitoring and automated decision-making, ranking top 3 among 150+ projects and demonstrating expertise in data analytics and Human Computer Interaction